Anoushkrit Goel

Researcher | **Data Scientist** | Entrepreneur | Artist
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Bio

Myself, **Anoushkrit Goel**, currently I am finishing up Masters by Research from **IIT Mandi** in Intelligent Systems under the guidance of Prof. Arnav Bhavsar and Prof. Aditya Nigam.
Here you will witness me working in different roles, as a **researcher**, an **entrepreneur** and as a **Data Scientist** with all but one vision to solve problems with my innovative and application-centric ideas. I thrive on open-ended problems with undefined paths.
However my obsession lies in building things, be it products, prototypes or research.

Education

Indian Institute of Technology Mandi, M.Tech by Research in *Intelligent Systems* Feb 2023- Present

- **GPA**: 9.47/10.0 *(Attended in-person)*
- **Coursework:** Deep Learning, Advanced Deep Learning, Biomedical Systems, Mobile Virtual Reality and AI, Research Methodology

Shiv Nadar University, B.Tech in Electronics and Communication Engineering Jul 2015 - May 2019

- **Major** : Electronics and Communication Engineering | **Minor**: Design |
- **GPA**: 7.61/10.0
- **Coursework**: Probability and Statistics, Data Structures and Algorithms, VLSI, Embedded Systems, Digital Systems etc.

Delhi Public School Ghaziabad, Class 12th | (Physics, Chemistry Maths) | 92.4% May 2015
Delhi Public School Ghaziabad, Class 10th | (Science Stream) | CGPA: 10 CGPA May 2013

Experience

6 + years of practical deep learning experience (4 years professional : 2.5 years research)
Researcher, *IIT Mandi*– Mandi, IN *(Physical)* current Feb 2023 - Feb 2025

- At IIT Mandi, my primary research is driven with an objective to create a end-to-end pipeline for Brain White Matter Analysis, which includes below mentioned projects **SlicerTracto**, (Slicer 3D extension for end-to-end brain MRI analysis) **TractoGPT**, (pretrained GPT on tractography streamline point cloud data) **TractoEmbed** (multi-embedding framework to learn representations of the point cloud data), **TractoAware** (multiple features other than (x,y,z) coordinates to segment point clouds using PointNet), and **Tract-RLformer** (GPT trained on tractography streamlines trajectories sampled through TD3 using Decision Transformer). [*Funded by SERB India*]. This end-to-end neuroimaging pipeline is in the process of being deployed at **PGI Chandigarh Hospital**.
- Published research papers in top conferences, refer *Publications* and shipped multiple AI projects with my collaborators like **MedIntel** (differential diagnosis medical llm), **VikramUnity**(Unity ml-agents based Chandrayaan Vikram Rover Simulation), and **Talk2Point**(in-progress) (multimodal generation and editing of 3D point cloud objects).

  - Point Cloud Networks, Transformers (GPT, BERT), LLM, pytorch, and Neuroimaging tools

Lead Data Scientist, *StatusNeo*– Gurgaon, IN *(Hybrid)* Aug 2022 - Mar 2023

- Led the Cortex AI Team for **Reliance Jio** which was part of BrainOS project, an AI system which spans across Jio's Verticals under the guidance of Dr. Shailesh Kumar and Akshay Mathur
- Focus of my team was to make a learning framework agnostic of current machine or deep learning frameworks with help of **protobufs** and **ONNX**.

Senior Data Scientist, *Gamezop* – Gurgaon, IN *(Remote)* Jun 2021 – Jun 2022

- Led the Deposit Prediction Project (**ClickVision**), where the objective was to predict "cash deposit" event on the platform within 5 mins of the sign-up user, to minimise churn and show ads or pop-ups to prospective users.
- This model reduced number of wasted offers and churning users, improving conversion rates to deposits. As

a ripple-effect UI changes were made to streamline the trajectory leading upto "cash deposit" on the platform. *refer **ClickVision** in Projects*

Founder & Chief Executive Officer, *Tensr.AI* – Noida, IN *(Remote)* Feb 2020 - Jun 2021

- Designing, prototyping, and building our proprietary device, Teresa Band, to capture the biomedical signals.
- Modelling of valvular malfunctions (mitral and tricuspid valves), and blood regurgitation using **Simulink** on **MATLAB**. Algorithms (**pyCardiogram**) to perform wave annotations to find correlations among different biomedical signals ECG, PPG and SCG.
- Hardware, Design, Software Dev, Business, and AI teams comprising a total of 15 team members with 3 co-founders.
- Brought government grants, established VC, and industry connections. Responsible for Data Science, Business Development, Design and Hardware teams.

Data Scientist, *xtLytics LLC* Noida, IN *(Physical)* Jan 2019 - Jan 2020

- CADtnet: Developed an edge-based AI app (**TFlite**) for the Low Middle Income Countries to Triage Breast Cancer using Ultrasound Images by clicking photos of the Ultrasound scans. [Medical Image]
- Invoice OCR using GCP and Tesseract which extracts and maps content in Tables (from images of tables) to a desired .csv, .xlsx format. [Image]

Publications

[IEEE ISBI'25] **TractoGPT: A GPT architecture for White Matter Tract Segmentation** **Mar 2025**
(accepted) https://arxiv.org/abs/2501.15464
IEEE 23th International Symposium on Biomedical Imaging, 2025
Anoushkrit Goel, Simroop Singh, Ankita Joshi, Ranjeet Ranjan Jha, Chirag Ahuja, Aditya Nigam, Arnav Bhavsar

[ICPR'24] **TractoEmbed: Modular Multi-level Embedding framework for white matter tract segmentation.** *(published)* **Dec 2024**
27th International Conference on Pattern Recognition, 2024
Anoushkrit Goel, Bipanjit Singh, Ankita Joshi, Ranjeet Ranjan Jha, Chirag Ahuja, Aditya Nigam, Arnav Bhavsar
https://link.springer.com/chapter/10.1007/978-3-031-78104-9_17

[ICPR'24] **Tract-RLFormer: A Tract-Specific RL policy based Decoder-only Transformer Network.** *(published)* **Dec 2024**
27th International Conference on Pattern Recognition, 2024
Ankita Joshi, Ashutosh Sharma, **Anoushkrit Goel**, Ranjeet Ranjan Jha, Chirag Ahuja, Arnav Bhavsar, Aditya Nigam
https://link.springer.com/chapter/10.1007/978-3-031-78201-5_17

[MICCAI '25] **TractRLFusion: A GPT-based RL policy fusion framework for tract-specific tractography** *(submitted)* **Dec 2024**
28th International Conference on Medical Image Computing and Computer Assisted Intervention

Projects

Detailing out projects done referred in **Experience**, most are private repos, request for access

SlicerTracto ongoing at IIT Mandi ***2024***
End-to-End Slicer 3D Extension for Neuroimaging tasks (Reconstruction, Tracking, Segmentation, Visualization, Metrics)

- SlicerTracto is an open-source Slicer 3D extension tool which enables Neuro-Radiologists to visualize, localize, track and segment white matter tracts from Diffusion MRI scans all in one stand-alone desktop application
- SlicerTracto runs smoothly as a local extension over Slicer on all devices, Mac, Windows and Linux. With an upcoming feature to run processes on cloud/servers.

MedIntel at IIT Mandi ***2024***
Differential diagnosis with fine-tuned med-llama

- MedIntel is a chat bot where we added the capability of performing differential diagnosis using **DDx-plus** dataset by fine-tuning MedLLaVA (by Microsoft) and med-llama (already fine-tuned on publicly-available MedQA and MedMCQA dataset, locally served on ollama.

TractoGPT at IIT Mandi ***2024***
TractoGPT is a general framework for learning streamlines data representation

- TractoGPT models Tractography streamlines (3D point sequences) as 3D point clouds where we create patches of point clouds from a bigger point cloud. Patches are encoded as tokens to pretrain GPT on point

cloud reconstruction task using dual-masking, which can further fine-tuned for downstream tasks like classification, segmentation etc.

- Outperformed Sherbrooke Connectivity Imaging Lab (SCIL)'s work, FIESTA and submitted at ISBI.

VikramUnity at IIT Mandi ***2023***

- A simulation of Chandrayaan Vikram Rover in Unity Game Engine where the rover is an agent and its objective is to carry out operations in daylight and reach targets with minimal energy expenditure and contours. Reward Policy is programmed accordingly, Rover and Terrain is also designed suited to the problem.

TractoEmbed at IIT Mandi ***2023***

Tractography Streamlines Segmentation using Point Cloud, multiple learnable embeddings from Streamline Data

- From point cloud representation of Tractography Streamlines we learn and extract multiple embeddings from 3 different data representations, self, regional and whole using a CNN, dVAE and PointNet respectively to classify into white matter anatomical tracts.
- Outperformed **Harvard Medical School's** work on Tract Segmentation, TractCloud.

TractoAware at IIT Mandi ***2022***

3D streamlines generated from the process of Tractography on diffusion MRI data)

- With point clouds as having having only (x,y,z) point information, we add information per point including normals, gradients, theta angle from the starting point of the streamline. With this we pass multiple features to point cloud networks to classify White Matter Tracts.

ClickVision at Gamezop ***2021***

Predicting clickstream events and scoring trajectories

- Inspired from MDP (Markov Decision Process) Model, we formulated the solution using transition matrices on clickstream data, where trajectories (sequence of clicks) leading up to the event gets more weightage, and importance of other actions/trajectories/nodes decay over iterations. A score is assigned to each trajectory based on its weight, discount factor and transition probabilities. With this we predict future actions of the user and calculate depositing score and give the verdict.

pyCardiograms at Tensr.AI ***2020***

Built upon NeuroKit, we extended the capabilities to add another modality SeismoCardiograms)

- Wave Analysis, Wave Annotation and Inter-wave correlation across ECG and SCG on PhysioBank Dataset.
- Prediction of SCG waveform for 2 heartbeat cycles using Bidirectional-LSTM on cardiac signals datasets.

Teresa Band at Tensr.AI ***2020***

Cardiovascular Triaging Wearable Device for Seismocardiograms)

- Teresa Band, a novel **Cardiovascular Triaging Wearable Device**, for detecting vascular and valvular cardiac conditions (like Arteriosclerosis, Turbulent Blood Flow) early, using mechanical vibrations of the heart, SCG (SeismoCardiogram), of the patient.

TableOCR at xtLytics ***2019***

In TableOCR, using Google Cloud Vision API and Tesseract we extracted text in different orientations and Parsing Costco Bills into tabular format (with key value pairs)

- TableOCR is for doing OCR on loosely Tabular Data, which is analogous to documents like menus, bills, invoices etc where there is implicit tabular information but lacks grid lines. For OCR, we used Tesseract and Cloud Vision API, and we match neighbouring bounding boxes on the basis of geometrical heuristics.

CADtnet: Computer-Aided Disease Triaging Network at xtLytics ***2019***

Breast Cancer Classification on Ultrasound Breast Scans deployed on Android Mobile using TFlite

- CADtnet is an edge DL app, for triaging Breast Ultrasound Scans based BI-RADS (Breast Imaging Reporting and Data System) scores which ranges from 0-6, where at a score of 3 app alerts the healthcare provider of suspected lesion with a bounding box surrounding the lesion. This app takes phone camera photos of Ultrasound Scans as input and is deployed in LMICs, and was tested in Guadalajara, Mexico.

Certifications and Courses

- Coursera & Deeplearning.ai's **Deep Learning Specialization** [WXBN3CDY3J33]
- DST DPIIT Startup India Certification for Tensr.AI Technologies Private Limited
- **NIDHI EIR** - SIIC IIT Kanpur, **Incubation & grant** - Catalyst IIT Mandi, **Pre-Incubation** at HTIC IIT Madras
- Rank 3 in Himalayan Startup Trek 2019, by Catalyst IIT Mandi

References

- **Shailesh Kumar** , Chief Data Scientist CoE AI/ML at Jio
- **Akshay Mathur**, Director of Technology, StatusNeo
- **Arnav Bhavsar**, Professor at IIT Mandi
- **Aditya Nigam** , Professor at IIT Mandi
- **Ranjeet Ranjan Jha**, Assistant Professor at IIT Patna
- **Harish Natarajan**, Director, HxCo India Private Ltd
- **Yashash Agarwal**, CEO and Founder at Gamezop
- **Upal Roy**, Associate Director - Advanced Analytics and Data Science at MSD (Merck & Co., Inc)

Ongoing Projects

- **Med3D**: Adding 3D multimodal input in the Medical LLMs, using Point-LLM for 3D context.
- **TractoSAM** Segmenting white matter tracts using SAM on 3D point cloud data.
- **Sculpt3D:** Fine-grained text-guided point cloud editing.
- **BiometricSEAL:** Embedding Biometric information into human generated images.
- **TumorGen:** Generalization across types of tumors which can segment new variants of tumors.
- **TumorProg:** Progression of tumors based on tumor proliferation rate and density.

Tech Stack

Languages: Python, **MATLAB**, C, C#, ProtoBuf, LateX
Tools in DL: PyTorch, ONNX, pandas, scikit-learn, scipy, wfdb, Gensim, SpaCy, numpy, plotly, Tensorboard, matplotlib, ollama, Open AI gym, Langchain, Tensorflow (Keras), Unity ML Agents, FB Prophet, SARIMA, neurokit, NLTK, PySpark, Open3D, OpenCV
For LLMs : Finetuning, Pretraining Language Models, 3D point cloud based LLMs, Tokenization Techniques for Multimodal LLMs, UnSloth, **RAG** (AdaptiveRAG, GraphRAG), Knowledge Graphs, Deep Reinforcement Learning (TD3, PPO, SAC), LangChain, LangGraph, AutoGen, DSPy
Frameworks: AWS, GCP, Simulink, Adobe Illustrator, Verilog, Slicer, MRtrix3, FSL, FreeSurfer, NeuroDesk, Unity, Power BI
Hobbies: Portrait Sketching, Wood Sculpting, Oil Painiting, Gym, Reading latest AI research